Summer Bootcamp
Session 1 – Linear Algebra

Matrices, linear systems, diagonalization, and statistical applications

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August 2026

Session Roadmap

Learning goals

By the end of the session, students should be able to:

- manipulate matrices and check dimension compatibility;
- write and solve linear systems in matrix form;
- interpret image, kernel, rank, invertibility, and transposition;
- compute eigenvalues and use diagonalization;
- connect symmetric matrices to covariance, portfolio variance, and PCA.

Section Roadmap

1. Algebraic Operations

Matrices

Definition

A matrix is a rectangular array of real numbers:

$$A = (a_{ij})_{1\leq i\leq n,\ 1\leq j\leq m} \in \mathcal{M}_{n,m}(\mathbb{R}).$$

It can be interpreted as a linear transformation from $\mathbb{R}^m$ to $\mathbb{R}^n$.

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Examples

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathcal{M}_{2,3}(\mathbb{R}), \qquad x = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} \in \mathbb{R}^3.$$

The dimensions match, so

$$Ax = \begin{pmatrix} 50 \\ 122 \end{pmatrix} \in \mathbb{R}^2.$$

Addition and Scalar Multiplication

Let $A = (a_{ij})$ and $B = (b_{ij})$ have the same size. Then

$$A + B = (a_{ij} + b_{ij}), \qquad \lambda A = (\lambda a_{ij}).$$

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Example

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 & -1 \\ -3 & 1 & 5 \end{pmatrix}.$$

$$A - 2B = \begin{pmatrix} -3 & -2 & 5 \\ 6 & 2 & -11 \end{pmatrix}.$$

Matrix Multiplication

If

$$A \in \mathcal{M}_{n,m}(\mathbb{R}), \qquad B \in \mathcal{M}_{m,p}(\mathbb{R}),$$

then $AB \in \mathcal{M}_{n,p}(\mathbb{R})$ and

$$(AB)_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}.$$

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Dimension checks

$$(2 \times 3)(3 \times 1) \to 2 \times 1, \qquad (4 \times 2)(2 \times 5) \to 4 \times 5.$$

$$(2 \times 3)(2 \times 5) \quad \text{is not defined.}$$

Matrix Multiplication: Worked Example

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 2 \end{pmatrix}.$$

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$$AB = \begin{pmatrix} 1 \cdot 2 + 2(-1) + (-1)3 & 1 \cdot 1 + 2 \cdot 0 + (-1)2 \\ 0 \cdot 2 + 3(-1) + 4 \cdot 3 & 0 \cdot 1 + 3 \cdot 0 + 4 \cdot 2 \end{pmatrix} = \begin{pmatrix} -3 & -1 \\ 9 & 8 \end{pmatrix}.$$

Order Matters

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

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$$\boxed{AB \neq BA}$$

Message

Matrix multiplication is associative and distributive, but not commutative in general.

Matrix-Vector Products

Let

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Interpretation

Each component of Ax is a dot product between a row of A and the vector x.

Section Roadmap

2 Associated Operators: Transpose, Trace, and Determinant

Transpose: Definition and Example

For $A = (a_{ij}) \in \mathcal{M}_{m,n}(\mathbb{R})$, the transpose $A^\top \in \mathcal{M}_{n,m}(\mathbb{R})$ is defined by

$$(A^\top)_{ij} = a_{ji}.$$

It turns rows into columns and columns into rows.

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$$\begin{pmatrix} 1 & 3 & -2 \\ 0 & 4 & 5 \end{pmatrix}^\top = \begin{pmatrix} 1 & 0 \\ 3 & 4 \\ -2 & 5 \end{pmatrix}.$$

Transpose Rules and Dot Products

Whenever dimensions are compatible,

$$(A+B)^\top = A^\top + B^\top, \qquad (\lambda A)^\top = \lambda A^\top, \qquad (AB)^\top = B^\top A^\top.$$

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For vectors $x, y \in \mathbb{R}^d$,

$$x^\top y = \sum_{i=1}^{d} x_i y_i, \qquad (Ax)^\top y = x^\top A^\top y.$$

Why the order reverses

The product AB means first apply B, then A. Transposition reverses this composition, hence $(AB)^\top = B^\top A^\top$.

Symmetric and Positive Semidefinite Matrices

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$$x^\top Ax \geq 0 \qquad \text{for every } x \in \mathbb{R}^n.$$

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For example,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad x^\top A x = (x_1 + x_2)^2 + x_1^2 + x_2^2 > 0 \quad (x \neq 0).$$

Why this matters

Covariance matrices are symmetric positive semidefinite, and portfolio variances are quadratic forms.

Trace: Definition and Main Rules

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The trace is linear and cyclic:

$$\operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B), \qquad \operatorname{tr}(\lambda A) = \lambda \operatorname{tr}(A), \qquad \operatorname{tr}(AB) = \operatorname{tr}(BA).$$

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For example,

$$\operatorname{tr}\begin{pmatrix} 3 & 1 & 0 \\ 2 & -4 & 5 \\ 1 & 1 & 7 \end{pmatrix} = 3 - 4 + 7 = 6.$$

Determinant: Meaning and the 2×2 Formula

The determinant measures signed volume scaling and detects dimension loss. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det(A) = ad - bc.$$

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$$A \text{ is invertible} \iff \det(A) \neq 0.$$

If $\det(A) = 0$, at least one direction is collapsed.

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If $\det(A) = 0$, at least one direction is collapsed.

$$\det \begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix} = -11, \qquad \det \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = 0.$$

The Horrible General Formula for the Determinant

For $A = (a_{ij}) \in \mathcal{M}_n(\mathbb{R})$,

$$\boxed{\det(A) = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}}.$$

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Here $\mathfrak{S}_n$ contains all permutations of $\{1, \ldots, n\}$ and

$$\operatorname{sgn}(\sigma) = (-1)^{N(\sigma)}, \quad N(\sigma) = \#\{(i,j) : i < j,\ \sigma(i) > \sigma(j)\}.$$

Computational warning

There are $n!$ signed products. The formula is fundamental for theory, but elimination or factorization is preferable in practice.

Laplace Expansion Along a Row or a Column

Delete row i and column j to obtain A_{ij}. Define

$$M_{ij} = \det(A_{ij}), \qquad C_{ij} = (-1)^{i+j} M_{ij}.$$

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$$\boxed{\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij}} = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(A_{ij}).$$

Expansion along a fixed column j gives

$$\boxed{\det(A) = \sum_{i=1}^{n} a_{ij} C_{ij}} = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det(A_{ij}).$$

Practical choice

Expand along a row or column containing many zeros.

Computing a 3×3 Determinant

For

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

expansion along the first row gives

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

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For example,

$$\det \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix} = 12 - 2(-6) = 24.$$

The nonzero determinant proves that the matrix is invertible.

Determinant Rules

For square matrices of the same size,

- $\det(AB) = \det(A)\det(B)$ and $\det(A^\top) = \det(A)$;
- $\det(A^{-1}) = 1/\det(A)$ when A is invertible;
- swapping two rows changes the sign;
- multiplying one row by λ multiplies the determinant by λ;
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For a triangular matrix,

$$\det(A) = \prod_{i=1}^{n} a_{ii}.$$

Section Roadmap

3. Inversion and Linear Systems

Identity and Inverse

Identity matrix

$$(I_n)_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

It satisfies $AI_n = A$ and $I_nA = A$ when dimensions are compatible.

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Inverse

A square matrix $A \in \mathcal{M}_n(\mathbb{R})$ is invertible if there exists A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

If A is invertible, the system $Ax = b$ has the unique solution $x = A^{-1}b$.

Inverse of a 2×2 Matrix

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the inverse exists exactly when $\det(A) = ad - bc \neq 0$. Then

$$\boxed{A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.}$$

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For the matrix used in the next example,

$$\begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix}^{-1} = \begin{pmatrix} 4/11 & 3/11 \\ 1/11 & -2/11 \end{pmatrix}.$$

Worked Example: Solving a Linear System

Consider the system $Ax = b$ with

$$A = \begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ -2 \end{pmatrix}, \qquad \det(A) = -11 \neq 0.$$

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Substitution gives

$$11x_2 = 9, \qquad x_2 = \frac{9}{11}, \qquad x_1 = -2 + 4x_2 = \frac{14}{11}.$$

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$$\boxed{x = \begin{pmatrix} 14/11 \\ 9/11 \end{pmatrix}}$$

Direct substitution and the inverse formula give the same solution.

Triangular Systems and Forward Substitution

Suppose

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \qquad Ly = \begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix}.$$

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The equations are

$$y_1 = P_1, \qquad y_1 + y_2 = P_2, \qquad y_1 + y_2 + y_3 = P_3.$$

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$$y_1 = P_1, \qquad y_2 = P_2 - P_1, \qquad y_3 = P_3 - P_2.$$

Finance Example: Discount Factors

Zero-coupon prices are observed:

$$P_1 = 0.97, \qquad P_2 = 0.92, \qquad P_3 = 0.855.$$

Write discount factors as partial sums:

$$D(T_1) = y_1, \quad D(T_2) = y_1 + y_2, \quad D(T_3) = y_1 + y_2 + y_3.$$

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$$y_1 = 0.97, \qquad y_2 = -0.05, \qquad y_3 = -0.065.$$

Section Roadmap

4. Injectivity, Surjectivity, Image, Kernel, and Rank

Image and Kernel

Let $A \in \mathcal{M}_{m,n}(\mathbb{R})$, viewed as a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Image / range

$$\mathrm{Im}(A) = \{Ax : x \in \mathbb{R}^n\} \subset \mathbb{R}^m.$$

It is the subspace reached by A.

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It is the set of directions collapsed by A.

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It is the set of directions collapsed by A.

Result – Rank–nullity theorem

The rank is $\operatorname{rank}(A) = \dim \operatorname{Im}(A)$, and

$$\operatorname{rank}(A) + \dim \ker(A) = n.$$

Invertibility Criteria

Result – Equivalent invertibility criteria

For $A \in \mathcal{M}_n(\mathbb{R})$, the following statements are equivalent:

- A is invertible;
- $\ker(A) = \{0\}$;
- $\mathrm{Im}(A) = \mathbb{R}^n$;
- $\mathrm{rank}(A) = n$;
- $Ax = b$ has a unique solution for every $b \in \mathbb{R}^n$.

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- $Ax = b$ has a unique solution for every $b \in \mathbb{R}^n$.

Practical warning

For numerical work, we usually solve $Ax = b$ directly instead of computing A^{-1} explicitly.

Orthogonality and Orthogonal Complements

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Two vectors $x, y \in \mathbb{R}^m$ are orthogonal when

$$x \perp y \quad \Longleftrightarrow \quad x^\top y = 0.$$

For a subspace $V \subset \mathbb{R}^m$, define

$$\boxed{V^\perp = \{y \in \mathbb{R}^m : y^\top v = 0 \text{ for every } v \in V\}.}$$

Geometric meaning

$V^\perp$ contains all directions perpendicular to every vector of V.

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For $V = \operatorname{span}\{(1,1,1)^\top\}$,

$$V^\perp = \{y \in \mathbb{R}^3 : y_1 + y_2 + y_3 = 0\}.$$

Image, Kernel, and Orthogonality

For $A \in \mathcal{M}_{m,n}(\mathbb{R})$,

$$\operatorname{Im}(A^\top) = (\ker A)^\perp, \qquad \operatorname{Im}(A) = (\ker A^\top)^\perp.$$

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Interpretation

The row space contains exactly the directions that are visible to the linear map. The kernel contains directions that disappear.

Example: Kernel and Image

Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

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Since $\dim \ker(A) = 1$ and A has three columns,

$$\operatorname{rank}(A) = 2.$$

Because A is symmetric, $\operatorname{Im}(A) = (\ker A)^{\perp}$ is the plane $y_1 + y_2 + y_3 = 0$.

Section Roadmap

5 Classical Decompositions
- Eigenvalue Diagonalization
- Spectral Decomposition
- Cholesky Decomposition

Eigenvalues and Eigenvectors

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Eigenvalues are found by solving

$$\det(A - \lambda I) = 0.$$

Diagonalization

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$$P = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}, \qquad D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

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Result – Powers of a diagonalizable matrix

$$A^k = PD^kP^{-1}, \qquad D^k = \operatorname{diag}(\lambda_1^k, \dots, \lambda_n^k).$$

Diagonalization: Algorithm

1. Compute the characteristic polynomial $p_A(\lambda) = \det(A - \lambda I)$.
2. Find the eigenvalues.
3. For each eigenvalue, solve $(A - \lambda I)v = 0$.
4. If there are n linearly independent eigenvectors, build P.
5. Write $A = PDP^{-1}$.

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3. For each eigenvalue, solve $(A - \lambda I)v = 0$.
4. If there are n linearly independent eigenvectors, build P.
5. Write $A = PDP^{-1}$.

Warning

Not every matrix is diagonalizable. A repeated eigenvalue may fail to provide enough independent eigenvectors.

Diagonalization: Worked Example

Let

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

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$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \qquad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}, \qquad A = PDP^{-1}.$$

Triangular Example for the Dynamics

For the dynamical application, consider

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}.$$

Its eigenvalues are the diagonal entries, with eigenvectors

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Hence

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \qquad A = PDP^{-1}.$$

Application: A Discrete-Time Dynamical System

Consider $x_{k+1} = Ax_k$ with

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}, \qquad x_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Using the previous diagonalization,

$$A^k = PD^kP^{-1} = \begin{pmatrix} 3^k & 3^k - 2^k \\ 0 & 2^k \end{pmatrix}.$$

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Therefore

$$x_k = A^k x_0 = \begin{pmatrix} 3^k - 2^k \\ 2^k \end{pmatrix}.$$

Interpretation

Diagonalization separates the dynamics into two scalar growth modes, 3^k and 2^k.

Symmetric Matrices: Spectral Theorem

Result – Spectral theorem

If A is real and symmetric, then:

- all eigenvalues are real;
- eigenvectors associated with distinct eigenvalues are orthogonal;
- there exists an orthogonal matrix Q and a diagonal matrix D such that

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Why this is central

Covariance matrices are symmetric. Their eigenvectors define orthogonal risk factors or principal components.

Application: Common and Spread Risk Factors

Consider the covariance matrix

$$C = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}.$$

Its spectral decomposition has

$$\lambda_1 = 1.8, \quad q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \lambda_2 = 0.2, \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

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The common position q_1 has variance 1.8, whereas the long–short position q_2 has variance 0.2.

Interpretation

The spectral theorem identifies orthogonal portfolios and quantifies the risk carried by each factor.

Cholesky Decomposition

Result – Cholesky decomposition

If A is symmetric positive definite, there is a unique lower triangular matrix L with positive diagonal entries such that

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Why this matters

Cholesky is used to solve positive-definite systems and to simulate correlated Gaussian vectors: if $Z \sim N(0, I)$ and $\Sigma = LL^\top$, then LZ has covariance Σ.

Application: Simulating Correlated Returns

Let Z_1, Z_2 be independent standard normal variables and set

$$L = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}, \qquad X = L \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = \begin{pmatrix} 2Z_1 \\ Z_1 + \sqrt{2} Z_2 \end{pmatrix}.$$

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Then

$$\mathrm{Cov}(X) = LL^\top = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}.$$

Thus X_1 and X_2 have variances 4 and 3, covariance 2, and correlation $2/\sqrt{12} = 1/\sqrt{3}$.

Practical use

Generate independent shocks first, then multiply by L to obtain shocks with the desired covariance.

Section Roadmap

6 Statistics and Finance Applications

Covariance Matrix from Data

Let $X \in \mathbb{R}^{n \times d}$ be a centered data matrix:

$$X = \begin{pmatrix} - & x_1^\top & - \\ - & x_2^\top & - \\ & \vdots & \\ - & x_n^\top & - \end{pmatrix}.$$

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The empirical covariance matrix is

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Result – Covariance matrices are positive semidefinite

$$\widehat{\Sigma}^\top = \widehat{\Sigma}, \qquad z^\top \widehat{\Sigma} z = \frac{1}{n-1}\|Xz\|^2 \geq 0.$$

Covariance Matrix: Numerical Example

Use the centered data matrix

$$X = \begin{pmatrix} 0.01 & -0.03 \\ 0.02 & 0.01 \\ -0.01 & 0 \end{pmatrix}.$$

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$$X = \begin{pmatrix} 0.01 & -0.03 \\ 0.02 & 0.01 \\ -0.01 & 0 \end{pmatrix}.$$

Then

$$X^\top X = \begin{pmatrix} 0.0006 & -0.0001 \\ -0.0001 & 0.0010 \end{pmatrix}, \qquad \widehat{\Sigma} = \frac{1}{n-1} X^\top X.$$

The matrix is symmetric and positive semidefinite by construction.

Portfolio Variance

Result – Portfolio variance formula

Let w be a vector of portfolio weights and Σ a covariance matrix of asset returns.

$$R_p = w^\top r, \qquad \boxed{\mathrm{Var}(R_p) = w^\top \Sigma w.}$$

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Example:

$$\Sigma = \begin{pmatrix} 0.04 & 0.018 \\ 0.018 & 0.09 \end{pmatrix}, \qquad w = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}.$$

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Example:

$$\Sigma = \begin{pmatrix} 0.04 & 0.018 \\ 0.018 & 0.09 \end{pmatrix}, \qquad w = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}.$$

$$w^\top \Sigma w = 0.6^2(0.04) + 2(0.6)(0.4)(0.018) + 0.4^2(0.09) = 0.03744.$$

PCA in One Slide

For centered data X, PCA diagonalizes

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Projection onto the first k components:

$$Z = XQ_k.$$

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Eigenvalues and principal directions:

$$\lambda_1 = 2, \quad q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \lambda_2 = 0, \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

All variance lies on the line $x_1 = x_2$.

Section Roadmap

From a Markov Chain to a Matrix

For states $\{1, \ldots, m\}$, use the column-vector convention

$$P_{ij} = \mathbb{P}(X_{n+1} = i \mid X_n = j).$$

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$$P_{ij} = \mathbb{P}(X_{n+1} = i \mid X_n = j).$$

Each column describes the next-state distribution from one current state:

$$P_{ij} \geq 0, \qquad \sum_{i=1}^{m} P_{ij} = 1.$$

Linear algebra link

A transition matrix is a nonnegative matrix that maps probability vectors to probability vectors.

Matrix Powers Represent Time

Result – Distribution dynamics

Let μ_n be the column vector containing the distribution of X_n. Then

$$\boxed{\mu_{n+1} = P\mu_n}, \qquad \boxed{\mu_n = P^n\mu_0}.$$

Matrix Powers Represent Time

Result – Distribution dynamics

Let μ_n be the column vector containing the distribution of X_n. Then

$$\boxed{\mu_{n+1} = P\mu_n}, \qquad \boxed{\mu_n = P^n\mu_0}.$$

The entry $(P^n)_{ij}$ is the probability of reaching state i after n steps when starting from state j.

Key message

Matrix multiplication composes transitions; taking the nth power advances the chain by n time steps.

Example: A Two-State Credit Chain

States are Good (G) and Bad (B), with

$$P = \begin{pmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{pmatrix}, \qquad \mu_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

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$$P = \begin{pmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{pmatrix}, \qquad \mu_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

After one and two years,

$$\mu_1 = P\mu_0 = \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix}, \qquad \mu_2 = P^2\mu_0 = \begin{pmatrix} 0.85 \\ 0.15 \end{pmatrix}.$$

Interpretation

Starting from Good, the two-year probability of being in Bad is 15%.

Stationary Distribution as an Eigenvector

Result – Stationarity condition

A stationary distribution satisfies

$$P\pi = \pi, \qquad 1^\top \pi = 1.$$

Thus π is an eigenvector for eigenvalue 1. For the credit chain,

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$$\pi = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}.$$

The other eigenvalue is 0.5, and for $\mu_0 = (1, 0)^\top$,

$$\mu_n = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} + 0.2(0.5)^n \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The stationary mode persists, while the transient mode decays geometrically.